Ian Marcus Breganza

Junior AI/ML Developer

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EXPERIENCE

Freelance AI/ML Developer *March 2026 – May 2026*

Real-Time Face Recognition Attendance System | Los Baños, Laguna

- Built and deployed an end-to-end face recognition attendance system on a Raspberry Pi 4, delivering a working production tool to a client that logs attendance automatically via live camera feed.
- Achieved 100% face detection rate (3,941/3,941 test images, 0 misses) by integrating a lightweight YOLOv5n-Face detector, and 96.5% face recognition accuracy (99.74% precision, 96.74% recall) across 20 individuals using ArcFace embeddings, validated under varied lighting (indoor, outdoor, low-light) and occlusion (with/without glasses) conditions.
- Diagnosed an inference latency bottleneck on Raspberry Pi 4 hardware; resolved it by swapping to the YOLOv5n-Face lightweight detection model and documented performance trade-offs (accuracy vs. speed) to guide the next system iteration.

IT & Systems Development Intern *Dec 2025 – Feb 2026*

PROMIS – Project Monitoring & Information System (Laravel / LAMP Stack) | DOST–FPRDI, UPLB

- Built the PNPKI digital certificate management module, managing PEM/CERT/P12 certificate files for ~50 employees with a dashboard tracking expiration dates and upload history, and automating signature insertion into generated PDF reports, eliminating manual signing across a 4-signatory approval chain for each quarterly report.
- Implemented password security enhancements for the system; presented the completed system to department heads and the Executive Committee.
- Designed and delivered a system onboarding workshop for ~50 senior stakeholders (department heads, executives, and scientists).

PROJECTS

Dipterocarp Species Classification via Dual-Branch Ensemble CNN *Apr 2024 – May 2026*

Thesis Project | National University Laguna, Calamba

- Designed and trained a dual-branch VGG16 CNN with a custom self-attention mechanism, fusing leaf venation and bark texture modalities to classify six Dipterocarp tree species from a 6,160-image field-collected dataset.
- Achieved 99.65% test accuracy (macro F1 0.9964) on the leaf branch and 99.62% test accuracy (macro F1 0.9965) on the bark branch, using two-stage fine-tuning with class-weighted loss to address dataset imbalance.
- Implemented an energy-based out-of-distribution (OOD) detection module calibrated via Youden's J-statistic, validated against domain-adjacent reference sets (BarkVN-50, Leaf Disease Dataset).
- Deployed the trained ensemble in a Tkinter desktop application for real-time field use by foresters; co-authored an IMRAD paper and an IEEE-formatted conference paper currently under submission, and led a 4-member team using Git for version control.

EDUCATION

Bachelor of Science in Computer Science *Expected July 2026*

National University, Laguna Campus

LANGUAGES & TECHNOLOGIES

ML / AI: PyTorch, TensorFlow, Keras, scikit-learn, spaCy, RoBERTa, NumPy, Pandas, OpenCV, YOLOv5n-Face, ArcFace

Languages: Python, PHP, Java, JavaScript, HTML/CSS

Web / Backend: Laravel (LAMP Stack), FastAPI, REST APIs, AJAX, OpenSSL / PNPKI

Tools: Git, Docker, VS Code, Kaggle, Raspberry Pi 4